

ENGR 857: Advanced Topics in Electrical and Computer Engineering Human-Robot Interaction

Spring 2026

San Francisco State University, School of Engineering

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Course Description

This interdisciplinary, project-driven course covers engineering, computational, and sociotechnical topics in the field of human-robot interaction (HRI) for real-world environments. Topics include research methods, robot system and interaction design, social signal processing (recognition and synthesis), and human-centered machine learning for safe and intuitive HRI. Students will learn foundational HRI research and concepts, design and implement a physical robot platform for real-time interaction with people, and create and conduct a study to evaluate their system.

Prerequisites: Graduate standing or permission of the instructor.

Course Objectives

By the end of this course, students should be able to:

- * Recognize key computational HRI topics including social signal processing, user modeling, and robot behavior adaptation. Be able to consider the strengths, limitations, and potential biases of each approach when applying these concepts to a given problem.
- * Connect current events with HRI and discuss their social and ethical implications.
- * Describe how system design influences human-centered factors including perception, usability, and trust of a robotic system.
- * Work together in teams to design, implement, and evaluate a robotic system with respect to both technical and human-focused factors.
- * Communicate technical research, methods, and results to a diverse audience in a written report and oral presentation.

Course Schedule

Course Component	Day and Time	Location
Lecture	Tuesday, Thursday 5:00 pm – 6:15 pm	SEIC 414
Office Hours	Wednesday 2:30 pm – 3:30 pm Thursday 4:00 pm – 5:00 pm	SEIC 326

Important Dates:

- * **Last day to Add, Drop, or Withdraw:** Monday, February 16
- * **Spring Break (no class):** Monday, March 23 – Saturday, March 28
- * **Cesar Chavez Day (no class):** Tuesday, March 31

Textbook and Resources

We will be referencing the HRI textbook (below), relevant literature from the field, and other resources. All lecture slides, supplementary materials, and assignments will be posted on Canvas.

1. Bartneck, C., Belpaeme, T., Eyssel, F., Kanda, T., Keijsers, M., & Sabanovic, S. (2024). *Human-Robot Interaction – An Introduction*. (2nd edition) Cambridge: Cambridge University Press.
Free PDF available here: <https://www.human-robot-interaction.org/>

If you are unfamiliar with Python, you may find the below resources to be helpful.

1. Beginner's Python Tutorial: Learn Python: <https://python.land/python-tutorial>
2. The Ultimate Python Beginner's Handbook: <https://www.freecodecamp.org/news/the-python-guide-for-beginners/>

Coursework and Grading

Course Grade

Each component will contribute to your final grade as follows:

Grading Component	Percentage
Assignments	30%
Class Participation	30%
Final Project Report and Presentation	30%
Final Peer Evaluations	10%

Final grades will be calculated according to the following scale:

Letter Grade	%	Letter Grade	%	Letter Grade	%	Letter Grade	%
		B+	87	C+	77	D+	67
A	93	B	83	C	73	D	63
A-	90	B-	80	C-	70	D-	60

Final grades will be truncated, **not** rounded. For example, an 82.8% would receive a B-.

Assignments

In this course, you will be working on and managing a project in teams. Thus, the assignments will be comprised of:

- Programming assignments
- Reading and reflections of relevant literature
- Group assignments

Late Assignments: Please check Canvas and the course calendar for updated due dates for deliverables to ensure that all assignments are submitted electronically by the due date. Late

assignments will lose 10% of the original points per day late. Many assignments are required for participation in in-class activities, so I highly encourage you to complete assignments by the posted due date. Penalty-free extensions for (unforeseeable) extenuating circumstances will be handled on a case-by-case basis, and only if you explain the situation at your earliest opportunity before the due date.

Regrades: Any request to regrade an item must be submitted within one week after receiving the grade. After one week, grades are final.

Class Participation

Class time will be spent in discussions and completing exercises which will contribute to your projects. You are expected to attend and actively participate in each class. Your ideas and contributions in class are valuable not only for your own learning, but also for your peers. Thus, please plan to arrive to class on time and prepared. Your participation grade will comprise on-time attendance in lecture and lab, participation with in-class activities, hours recorded in weekly work logs, and evaluations from your team members.

Absences: In the event that you are unable to attend class in person (e.g., if you are sick or traveling), all content and assignments will be posted to Canvas by the end of the day. However, it is your responsibility to find out what you missed (e.g., check with your classmates regarding information or announcements). You are also expected to work with your team to ensure that all group work gets completed.

I will excuse up to **three absences**. Penalty-free extensions for (unforeseeable) extenuating circumstances will be handled on a case-by-case basis, and only if you explain the situation at your earliest opportunity before the missed class. Other absences will result in no participation points for that day. Please note that you are still responsible for working with your team to complete any in-class group work for the day. **Please do not come to class if you feel unwell.**

Final Project Report and Presentation

This course centers around a project which you will work on throughout the semester. You will work in groups of ~4 on a project. All team members are expected to actively contribute to the project, including programming, documentation, final report, and final presentation. More details will be provided later in the semester.

Class Policies

Collaboration Policy

Collaboration is encouraged. You may discuss approaches to completing assignments and solving problems with anyone in the class, including the professor. However, **copying** solutions from any source (including but not limited to other people, books, websites, and ChatGPT) is prohibited and will be treated as an academic integrity violation. Always cite any people, resources, or software that you reference when completing any assignment.

Individual and pair programming (i.e., working with one other student) is allowed for all programming assignments. In pair programming, it is expected that both students will participate fully for each part of all labs. Both students should submit the assignment to Canvas and indicate who you worked with on the submission.

All students are expected to complete assignments in accordance with the university's Academic Integrity policy (see below). You are expected to be honest and ethical in your academic work. Cheating and plagiarism are serious violations of the academic code of conduct. There is a zero-tolerance policy for cheating in this class. Any assignment or exam that is the product of cheating will receive zero credit, and you may be reported to the Student Judiciary Affairs.

Policy Regarding Generative AI and AI Tools

During our class, **we may use Generative AI as a tool like working with a classmate on an assignment.** While you can “collaborate” (e.g., “discuss” questions you have, clarify topics), the final product should be created by you. If you use generative AI, you will be required to include proper attribution in your submitted assignment, including the prompts you used or a link to your chat history.

Generative AI tools are not a replacement for creativity, originality, and critical thinking, and you will be held fully responsible for any work submitted. Any use outside of this permission will be considered a violation of the university's Academic Integrity policy (see below), and you may be reported to the Student Judiciary Affairs.

SF State Policies

Academic Integrity

Academic integrity, the ethical presentation of one's own work in accordance with the rules established for this class, is required. Instances of academic misconduct will be reported to the College in which the course is housed, the Division of Graduate Studies (if a graduate student),

and the Office of Student Conduct with the report being kept in those offices until a student earns his/her degree. Any instances of cheating, deceit, fabrication, forgery, plagiarism, unauthorized altering of records or submitting false documents, unauthorized collaboration, unauthorized submission of work previously given credit, or other forms of academic misconduct will be assigned a grade penalty, likely an F or a grade of zero. Failing one or more assignments or examinations for reasons of academic integrity violations may result in a final class grade of F. Students may not withdraw from classes in which they have committed academic misconduct. Consequences for violations of academic integrity may exceed an F on the assignment, examination, or class as determined by the Academic Integrity Review Committee.

Members of our academic community have a responsibility to develop an awareness of academic integrity, to cultivate skills to realize honesty in academic and community work, and to sustain actively academic honor as a core value of our community. Students are expected to engage in behaviors that reflect well upon the university. In addition to attending to one's own actions, the Standards for Student Conduct require that students who witness academic dishonesty notify their faculty/instructor, department chair, or the Office of Student Conduct. Supporting academic integrity enhances the reputation of the University and the value attributed to degrees awarded by the University. For more detailed information, visit: <http://conduct.sfsu.edu/plagiarism>.

Disability Access

Students with disabilities who need reasonable accommodations are encouraged to contact the instructor. The Disability Programs and Resource Center (DPRC) is available to facilitate the reasonable accommodations process. The DPRC is located in the Student Service Building and can be reached by telephone (voice/415-338-2472, video phone/415-335-7210) or by email (dprc@sfsu.edu).

Disclosures of Sexual Violence (Title IX)

SF State fosters a campus free of sexual violence including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the Title IX Coordinator by completing the report form available at <https://titleix.sfsu.edu>, emailing [ypsaeam@sfsu.edu](mailto:ypsaem@sfsu.edu), or calling (415) 338-2032. To disclose any such violence confidentially, contact:

- * **The SAFE Place** – (415) 338-2208; http://www.sfsu.edu/~safe_plc/
- * **Counseling and Psychological Services Center** – (415) 338-2208; psyservs.sfsu.edu

For more information on your rights and available resources, visit <http://titleix.sfsu.edu>.

Policy on Observance of Religious Holidays

If a student wishes to observe religious holidays and such observances require the student to be absent from class activities, it is the responsibility of the student to inform the instructor, in writing, about such holidays during the first two weeks of the class each semester. If such holidays occur during the first two weeks of the semester, the student must notify the instructor, in writing, at least three days before the date that he/she will be absent. Please check <http://www.interfaith-calendar.org/> for world religious days.

Land Acknowledgement

The campuses of San Francisco State University on the San Francisco Peninsula and North Bay are located within the occupied territories of the Ramaytush Ohlone (Rah-my-toosh O-lone-ee) and the Coastal Miwok peoples (who, along with the Southern Pomo, are organized as the Federated Indians of Graton Rancheria). As we discuss the interconnectedness of technology, culture, and community, our learning and practice must be rooted in the process of decolonization and the recognition of the many Indigenous cultures who continue to uphold their relations and stewardship of these lands.